

Multi-State High School Robotics Competition
Period of Performance: March 18, 2026 to June 30, 2026

A. Background

NASA Glenn Research provides resources and support for a multi-state high school robotics competition to be held in Cleveland, Ohio each year. The competition provides high-school students with the opportunity to work alongside engineers and other skilled technical mentors to design, build, program, and compete in challenges with sophisticated robots. In January, a live kickoff event is to be held during which the teams are given a robotics challenge for the year. The teams then have six weeks to utilize the standard kit of parts and their own materials to build a robot which is able to accomplish the challenge of the game.

B. Goals

The robotics competition engages students by encouraging them to become excited about science, technology, and skilled technical trades in a supportive, collaborative team environment equips students with both knowledge and skills needed to enter the aerospace workforce.

The robotics competition is strongly aligned to NASA's agency-wide priorities for STEM. Specifically, OSTEM's goal to build the future skilled technical aerospace workforce. The competition helps students transition to work in the skilled technical disciplines. Through this active learning experience, high-school students learn what academic skills are required to succeed and participate in an exciting hands-on experience similar to skilled technical roles in aerospace workplaces.

Students participating in the robotics competition gain:

- Authentic, real-world problem solving through hands-on skill development,
- Discovery of skills and approaches they will see in Industry through mentor interactions,
- Work-based learning through simulated job roles including skilled technical trades, where they approach problems to make an impact for the team and their community,
- Access to industry mentors that guide students in their learning, and
- Understanding of how their learning applies to student career pathways.

C. Task(s) and subtasks

The funding for this grant will be used to support the multi-state robotics competition to be held in Cleveland, Ohio. The funding provides skilled technical aerospace workforce building opportunities for NASA, encourages NASA's engineers and skilled technical professionals to participate as Mentors for high school robotics teams, and encourages participation at the competition.

NASA provided grant funds shall be used toward the cost of operating the multistate high school robotics competition at the Cleveland State University Wolstein Center. In support

of this grant, the awardee shall use the \$25,000 grant funds for venue rent and services (e.g., event staff, police, fire & safety, first aid, EMT transport, security, cleaning, arena conversion, operations labor, utility surcharge, electric outlet charge, and front parking lot). Additional venue charges and competition costs will be paid for by the vendor or other non-NASA funding sources.

D. Deliverables

1. Conduct the multi-state high school robotics competition at Cleveland State University Wolstein Center.
2. Financial Reporting: Provided a detailed financial report listing all expenditures of grant funds no later than 120 days after the end of the period of performance.
3. Final Report: Provide a detailed report the outcomes of the multi-state high school robotics competition including at a minimum:
 - a. Project activities and accomplishments measured against proposed goals and objectives,
 - b. Evidence of how the project activities furthered stakeholder priorities,
 - c. Extent to which stakeholder collaborations have evolved, and
 - d. Participation metrics.

E. POCs:

Project Manager

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